SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

Annexure A

APPLICATION-BASED QUESTIONS

Code of Conduct:

*I **Rithish Kumar D (192565089)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

APPLICATION-BASED QUESTIONS

Cryptography and Network Security — CSA51 | CO1

Questions Answered in this Submission: 2 & 3

Q2 — Intercepted Suspicious Message · Q3 — Secure Messaging System (Keyword-Based Encryption)

QUESTION 2

Intercepted Suspicious Message

Scenario: A cybersecurity intern intercepts the coded message “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Question

(a) Deduce the most likely original message.

(b) Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Approach: Reading the Clues

Short word repeated twice
“RH” appears as its own word twice — in English, the most common repeated 2-letter word is “IS.”

Single-letter word “Z”
The only common single-letter English words are “A” and “I” — “Z” most likely maps to “A.”

No frequency analysis needed
Only 4 distinct words, all short — consistent with a simple monoalphabetic (one letter always maps to one other letter) substitution.

Test a mirror-alphabet rule
Classical ciphers often mirror the alphabet (A↔Z, B↔Y …); worth testing directly against the ciphertext.

QUESTION 2 · PART A (continued)

Decoding the Message — Atbash (Mirror) Substitution

Testing the mirror-alphabet rule: each ciphertext letter is replaced by the letter the same distance from the opposite end of the alphabet (A↔Z, B↔Y, C↔X …).

Ciphertext	G	S	R	H	(space)	R	H	(space)	Z	(space)	H	V	X	I	V	G
Plaintext	T	H	I	S	(space)	I	S	(space)	A	(space)	S	E	C	R	E	T

Decoded message: “THIS IS A SECRET”

Proof: the Atbash Rule Holds for Every Letter

For every letter pair, position-in-alphabet(cipher) + position-in-alphabet(plain) = 27 — the signature of an Atbash (mirror) substitution.

Cipher	Position	Plain	Position	Sum
G	7	T	20	27
S	19	H	8	27
R	18	I	9	27
H	8	S	19	27
Z	26	A	1	27
V	22	E	5	27
X	24	C	3	27

Conclusion

Every pair sums to 27, confirming this is a classical Atbash cipher — not an arbitrary substitution.

QUESTION 2 · PART B

Justification — Why This Reading Is Correct

Consistent one-to-one mapping

Every occurrence of a ciphertext letter decodes to the same plaintext letter throughout the message (e.g., 'R' always becomes 'I', 'H' always becomes 'S') — the defining property of a monoalphabetic substitution cipher.

Repeating short-word pattern

"RH" occurs twice as an isolated word and decodes both times to "IS", matching natural English usage ("this IS a secret") rather than coincidence.

Plausible word lengths and structure

Word lengths (4, 2, 1, 6 letters) map onto common English words — THIS, IS, A, SECRET — forming a grammatically valid sentence.

Mathematical signature of Atbash

Every letter pair's alphabet positions sum to 27 (n + 1, where n = 26 letters), which is the exact identity of the Atbash cipher — confirming the mechanism, not just guessing the answer.

Fits the scenario

The scenario states the message uses "a simple substitution mechanism rather than complex encryption" — Atbash, a fixed, key-less mirror substitution, matches that description precisely.

QUESTION 3

Secure Messaging System

Scenario: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Question

(a) Demonstrate how the encryption process transforms the message step-by-step.

(b) Explain how the repeating keyword influences each character of the ciphertext.

Identifying the Technique

A “keyword-based” cipher that shifts each plaintext letter by a repeating key is the Vigenère cipher — a polyalphabetic substitution. Each letter is assigned a numeric value (A = 0, B = 1, …, Z = 25), and ciphertext = (plaintext value + key value) mod 26.

Step 1 — Align the Repeating Keyword

The 3-letter keyword “KEY” is repeated cyclically to match the length of “HELLO” (5 letters):

Plaintext	H	E	L	L	O
Key (repeated)	K	E	Y	K	E

QUESTION 3 · PART A (continued)

Step-by-Step Vigenère Encryption

Formula: Ciphertext letter = (Plaintext value + Key value) mod 26, using A = 0, B = 1, …, Z = 25.

Plaintext	P value	Key letter	K value	P + K	mod 26	Ciphertext
H	7	K	10	17	17	R
E	4	E	4	8	8	I
L	11	Y	24	35	9	J
L	11	K	10	21	21	V
O	14	E	4	18	18	S

Resulting ciphertext: **“RIJVS”**

Worked Example — First Letter

H (7) + K (10) = 17 → 17 mod 26 = 17 → the 17th letter (0-indexed) is ‘R’. The same arithmetic is repeated for every letter, using whichever key letter aligns with that position.

QUESTION 3 · PART B

How the Repeating Keyword Shapes Each Character

Cyclic alignment

Because “KEY” (3 letters) is shorter than “HELLO” (5 letters), it wraps around: positions 1–3 use K, E, Y and positions 4–5 reuse K, E. Each plaintext letter is shifted by a different key letter depending on its position, not a single fixed shift.

Same plaintext letter → different ciphertext letters

‘L’ appears twice in HELLO (positions 3 and 4), but is paired with different key letters (‘Y’ then ‘K’), so it encrypts to two different ciphertext letters: ‘J’ and ‘V’. A simple Caesar shift could never do this.

Polyalphabetic effect

Each key letter effectively selects a different Caesar-shifted alphabet for that position — K shifts by 10, E shifts by 4, Y shifts by 24 — so the message is encrypted through multiple alphabets rather than one.

Security implication

Because identical plaintext letters can map to different ciphertext letters, the ciphertext's letter-frequency pattern is flattened. This makes the Vigenère cipher far more resistant to simple frequency analysis than the Atbash/monoalphabetic cipher used in Question 2 — though a repeating key of known length is still recoverable via Kasiski examination or the index of coincidence.